

Counting with Mobius Function

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1 Introduction

Mobius function is often introduced out of nowhere in the when students learn about arithmetic functions in a typical number theory course. This expository paper explains the motivation of defining the Mobius function (namely, the inverse of a particular element in Dirichlet ring of arithmetic functions) and presents two applications of mobius inversion to counting problems.

2 Dirichlet Convolution

A Dirichlet series is any series of the form.

$$\sum_{n=1}^{\infty} \frac{a_n}{n^x},$$

The n^{th} -power term of

$$\left(\sum_{n=1}^{\infty} \frac{a_n}{n^x}\right) \cdot \left(\sum_{n=1}^{\infty} \frac{b_n}{n^x}\right)$$

is given by

$$\sum_{d|n} a_d b_{n/d} \frac{1}{n^x}$$

The n^{th} coefficients in the product is found by summing product of all pairs of coefficients in each series whose indexes multiply to be n . This is highly analogous to convolution of continuous functions, so it is natural to generalize convolution to arithmetic functions.

Definition 2.1. *An arithmetic function is a function $f : \mathbb{N} \rightarrow \mathbb{C}$* ¹

Every arithmetic function can be encoded with a sequence in which the n^{th} entry is equal to $f(n)$. We write

$$f = (f_1, f_2, f_3, \dots)$$

Definition 2.2. *The Dirichlet convolution $*$ is a binary operation defined by:*

$$f * g(n) = \sum_{d|n} f(d)g\left(\frac{n}{d}\right)$$

¹The most general definition is that the codomain is $\mathbb{C}$, but for our purposes $\mathbb{N}$ is enough.

The set of arithmetic functions, equipped with Dirichlet convolution, forms an abelian group. We now introduce a few important elements of this group.

Definition 2.3. The constant one function, denoted $\bar{1}$, is defined by

$$\bar{1}(n) = 1$$

for every positive integer n . As a sequence,

$$\bar{1} = (1, 1, 1, 1, \dots).$$

Definition 2.4. The epsilon function is defined by

$$\epsilon(n) = \begin{cases} 1, & n = 1, \\ 0, & n > 1. \end{cases}$$

Equivalently,

$$\epsilon = (1, 0, 0, 0, \dots).$$

The epsilon function plays the role of the identity element.

Theorem 2.5. For every arithmetic function f ,

$$f * \epsilon = \epsilon * f = f.$$

Proof. For every positive integer n ,

$$(f * \epsilon)(n) = \sum_{d|n} f(d) \epsilon\left(\frac{n}{d}\right).$$

Since $\epsilon(m) = 0$ unless $m = 1$, the only nonzero term in the sum occurs when

$$\frac{n}{d} = 1,$$

that is, when $d = n$. Hence

$$(f * \epsilon)(n) = f(n)\epsilon(1) = f(n).$$

The proof that $\epsilon * f = f$ is identical. □

Dirichlet convolution naturally appears in many familiar arithmetic functions.

Let

$$\text{id}(n) = n.$$

Then

$$(\text{id} * \bar{1})(n) = \sum_{d|n} d = \sigma(n),$$

where $\sigma(n)$ denotes the sum of the positive divisors of n .

Since

$$\bar{1}(d)\bar{1}(n/d) = 1$$

for every divisor d of n ,

$$(\bar{1} * \bar{1})(n) = \sum_{d|n} 1 = \tau(n),$$

where $\tau(n)$ is the number of positive divisors of n .

Theorem 2.6. *The inverse of $\bar{1}$ is*

$$\mu(n) = \begin{cases} 1 & \text{if } n = 1, \\ (-1)^k & \text{if } n \text{ is square-free with } k \text{ distinct prime factors,} \\ 0 & \text{if } n \text{ has a squared prime factor.} \end{cases}$$

Proof. By definition of inverse, $\bar{1} * \mu = \epsilon$. So

$$(1, 1, 1, 1, \dots) * (\mu_1, \mu_2, \mu_3, \dots) = (1, 0, 0, 0, \dots)$$

from which we obtain, for every $n > 1$,

$$\sum_{d|n} \mu(d) = 0,$$

while

$$\mu(1) = 1.$$

For example,

$$\begin{aligned} \mu(1) &= 1, \\ \mu(1) + \mu(2) &= 0, \\ \mu(1) + \mu(3) &= 0, \\ \mu(1) + \mu(2) + \mu(4) &= 0, \\ &\dots \end{aligned}$$

Consider the Hasse diagram of the divisibility posets of n .

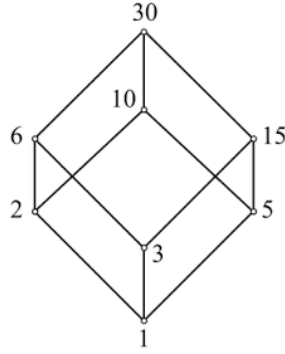


Figure 1: Hasse diagram of the divisors of 30

Suppose $n = p_1 p_2 \cdots p_k$ is square-free. Every divisor of n corresponds to choosing a subset of its prime factors, so every divisor appears exactly once in the Hasse diagram. Since every proper divisor has already had its value determined, the equation

$$\sum_{d|n} \mu(d) = 0$$

determines $\mu(n)$.

We prove by induction on k that

$$\mu(n) = (-1)^k.$$

The cases $k = 0$ and $k = 1$ are immediate. Since $\mu(1) + \mu(p) = 0$, it must be that $\mu(p) = -1$. Assume the statement holds for every square-free number having fewer than k distinct prime factors. Then

$$0 = \sum_{d|n} \mu(d) = \mu(n) + \sum_{S \subsetneq \{1, \dots, k\}} (-1)^{|S|},$$

where each proper divisor corresponds to a proper subset S of the prime factors.

Using the binomial identity,

$$\sum_{i=0}^k \binom{k}{i} (-1)^i = (1 - 1)^k = 0,$$

we obtain

$$\sum_{S \subsetneq \{1, \dots, k\}} (-1)^{|S|} = -(-1)^k.$$

Hence

$$\mu(n) = (-1)^k.$$

Now suppose n is not square-free. As an example, consider the divisor lattice of 24.

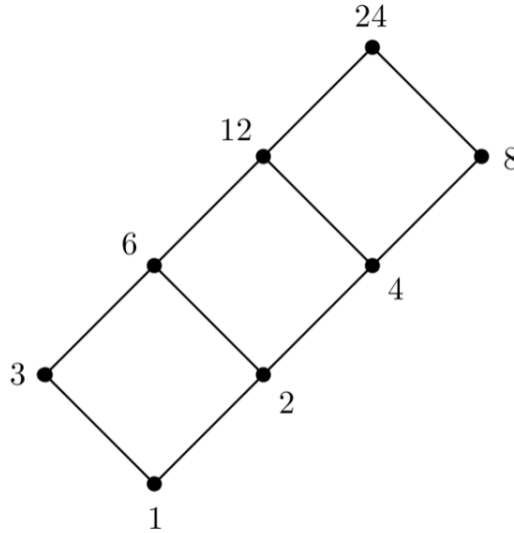


Figure 2: Hasse diagram of the divisors of 24

The divisors pair up according to whether they contain one copy of the repeated prime or two copies. Every square-free divisor contributes the opposite value of its corresponding divisor containing the extra copy of the repeated prime. Consequently, every proper divisor cancels with another proper

divisor.

Therefore,

$$\sum_{\substack{d|n \\ d < n}} \mu(d) = 0,$$

forcing

$$\mu(n) = 0.$$

Combining the two cases gives the desired formula for the inverse of $\bar{1}$. □

Theorem 2.7 (Möbius Inversion). *Suppose arithmetic functions f and g satisfy*

$$g(n) = \sum_{d|n} f(d).$$

Then

$$f(n) = \sum_{d|n} \mu(d) g\left(\frac{n}{d}\right).$$

3 Applications to Counting

3.1 Primitive Necklaces

Suppose we have k colors available and wish to color a circular necklace with n beads. Two colorings are considered the same if one is obtained from the other by rotation.

A necklace is called *primitive* if it cannot be obtained by repeating a shorter necklace multiple times.

Let $M(n)$ denote the number of primitive necklaces with n beads. Instead of counting primitive necklaces directly, it is easier to count all strings of length n . There are exactly k^n such strings.

Every string is obtained uniquely by repeating a primitive necklace whose length divides n . Furthermore, each primitive necklace of length d gives rise to exactly d distinct strings, corresponding to its d cyclic rotations.

Therefore,

$$k^n = \sum_{d|n} d M(d).$$

Applying Möbius inversion gives

$$nM(n) = \sum_{d|n} \mu(d) k^{n/d},$$

$$M(n) = \frac{1}{n} \sum_{d|n} \mu(d) k^{n/d}$$

If we wish to count all necklaces, primitive or not, let $N(n)$ denote this number. Since every necklace is obtained by repeating a primitive one,

$$N(n) = \sum_{d|n} M(d),$$

and it's not hard to see that

$$N(n) = \frac{1}{n} \sum_{d|n} \varphi(d) k^{n/d}.$$

3.2 Monic Irreducible Polynomials

Let $\mathbb{F}_q$ be the finite field with q elements, and let $I(n)$ denote the number of monic irreducible polynomials of degree n over $\mathbb{F}_q$.

Theorem 3.1.

$$I(n) = \frac{1}{n} \sum_{d|n} \mu(d) q^{n/d}.$$

The key observation is that every element of the extension field $\mathbb{F}_{q^n}$ has a unique minimal polynomial over $\mathbb{F}_q$. We first establish three lemmas.

Lemma 3.2. *Over $\mathbb{F}_{q^n}$,*

$$x^{q^n} - x = \prod_{a \in \mathbb{F}_{q^n}} (x - a).$$

Proof. The multiplicative group $\mathbb{F}_{q^n}^\times$ has order $q^n - 1$. Hence every nonzero element satisfies

$$a^{q^n - 1} = 1,$$

so

$$a^{q^n} = a.$$

$a = 0$ also satisfies the equation, so every element of $\mathbb{F}_{q^n}$ is a root of $x^{q^n} - x$. Since there are exactly q^n distinct roots, the polynomial factors as claimed. $\square$

Lemma 3.3. *Let $m(x)$ be an irreducible polynomial over $\mathbb{F}_q$ of degree d . Then*

$$m(x) \mid x^{q^n} - x \iff d \mid n.$$

Proof. We omit the proof, because surely one can find it in algebra or number theory textbooks! $\square$

Lemma 3.4. *Every irreducible polynomial over a finite field is separable. Consequently, distinct irreducible polynomials have no common root.*

Proof. This is a standard result from finite field theory, we shall use it without proof. $\square$

²This polynomial in k is sometimes called the *necklace polynomial*.

By Lemmas 2 and 3,

$$x^{q^n} - x$$

is the product of all monic irreducible polynomials whose degrees divide n . Taking degrees of both sides,

$$q^n = \sum_{d|n} d I(d).$$

This identity is formally identical to the one obtained for primitive necklaces! Applying Möbius inversion yields

$$I(n) = \frac{1}{n} \sum_{d|n} \mu(d) q^{n/d}.$$

It is remarkable that two seemingly unrelated counting problems satisfy the same convolution identity, and therefore share the same Möbius inversion formula.

References

- [1] Kenneth Ireland and Michael Rosen, *A Classical Introduction to Modern Number Theory*, 2nd ed., Graduate Texts in Mathematics, Vol. 84, Springer-Verlag, New York, 1990.
- [2] Álvaro Lozano-Robledo, “Counting irreducible polynomials,” *Mathematics Stack Exchange*, Dec. 19, 2011. Available at: <https://math.stackexchange.com/questions/92747/counting-irreducible-polynomials> (Accessed July 31, 2026).